Sarah Chen

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Summary

Machine Learning Engineer and NLP Researcher with 5 years building production ML systems. Specialised in large language models, RAG pipelines, and embedding-based semantic retrieval. Published 2 papers at ACL. PhD in Computational Linguistics from University of Washington.

Technical Skills

- ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, sentence-transformers
- NLP: BERT, GPT, RAG pipelines, Named Entity Recognition, semantic search, embeddings
- Vector Databases: ChromaDB, Pinecone, FAISS, Weaviate
- Languages: Python (expert), R, SQL, Bash
- MLOps: MLflow, DVC, Weights & Biases, Docker, Kubernetes
- Cloud: AWS SageMaker, GCP Vertex AI, Azure ML

Work Experience

Senior ML Engineer — Amazon Alexa AI, Seattle (Jun 2021 – Present)

- Built multilingual intent classification system (15 languages) deployed to 300 million Alexa devices.
- Designed RAG pipeline for Alexa Q&A; using sentence-transformers + DynamoDB; improved accuracy by 18%.
- Reduced model inference latency 3× through ONNX quantisation and TensorRT optimisation.

ML Engineer — Semantic Health, Seattle (Sep 2018 – May 2021)

- Developed clinical NLP pipeline extracting 40+ entity types from medical notes using BioBERT.
- Fine-tuned GPT-2 for clinical text summarisation; saved 3 hours per physician per week.

Education

Ph.D. Computational Linguistics — University of Washington (2019) | GPA: 3.95/4.0

B.Sc. Computer Science — MIT (2014) | GPA: 4.0/4.0

Publications

- 'Cross-lingual Retrieval for Low-Resource Languages' — ACL 2022
- 'Efficient Semantic Search with Compressed Embeddings' — EMNLP 2020